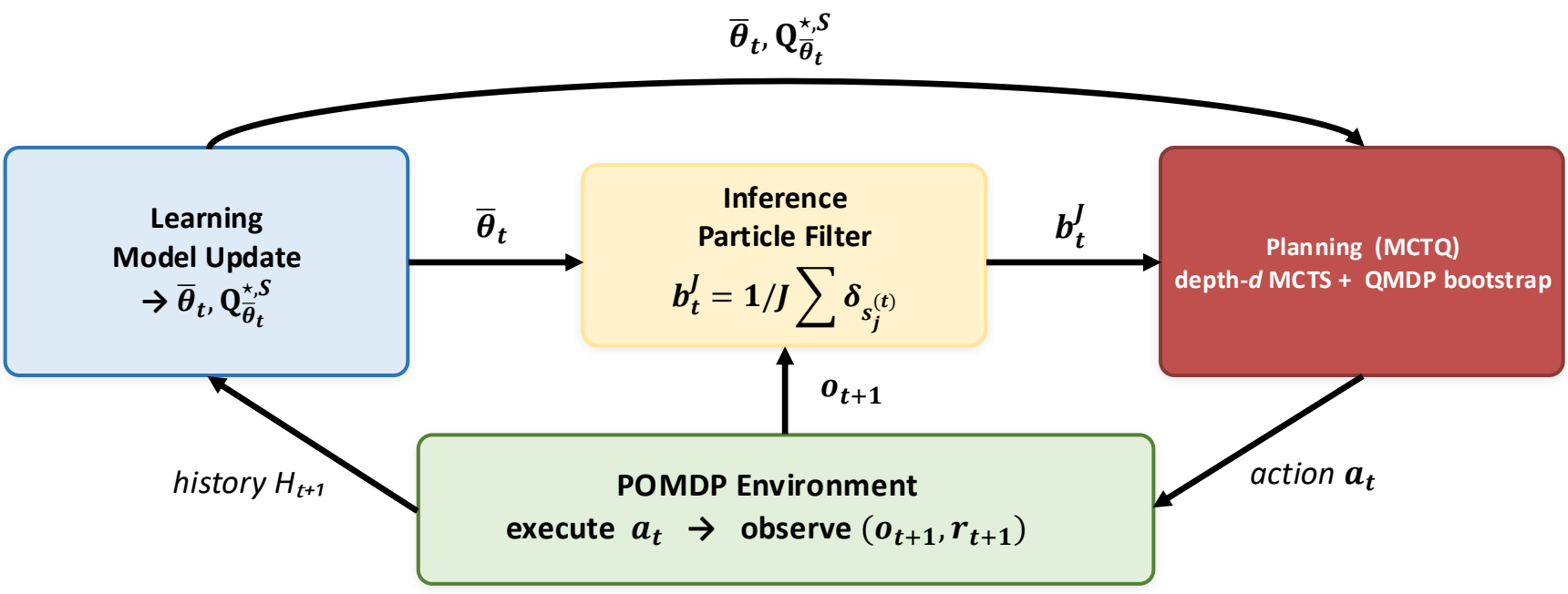

$\overline{\theta}_t, Q^{\star,S}_{\overline{\theta}_t}$
Learning
Model Update
$\rightarrow \overline{\theta}_t, Q^{\star,S}_{\overline{\theta}_t}$
$\overline{\theta}_t$
Inference
Particle Filter
$b^J_t = 1/J \sum \delta_{s^{(t)}_j}$
$b^J_t$
Planning (MCTQ)
depth-$d$ MCTS + QMDP bootstrap
$o_{t+1}$
history $H_{t+1}$
POMDP Environment
execute $a_t$ $\rightarrow$ observe $(o_{t+1}, r_{t+1})$
action $a_t$